Master's Thesis

Abdominal Multi-Organ Segmentation Based on Feature Pyramid Network and Spatial Recurrent Neural Network

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Abstract

After entering the 21st century, many countries are facing the challenge of population aging. One of the most important tasks brought by population aging is that senior citizens need timely and continuous health monitoring. To ensure that senior citizens can get proper medical support, there are two primary aspects that social welfare investment should focus on.

On one hand, physical inspection devices should be deployed as widely as possible to improve the accessibility for senior citizens and other people in need. For example, the caring centers can maintain some inspection devices for regular physical checks like glucometers and sphygmomanometers. Besides those comprehensive and relatively naive health indicators like blood glucose or pressure, other inspection methods are needed as well to analyze more health information. Ultrasound imaging is one of the most important medical imaging modalities. Because it is approachable, effective, informative and low cost, ultrasound devices can be easily implemented. Also benefiting from its non-invasive and non-radioactive nature, the operation of ultrasound devices is of low threshold. Ultrasound imaging can provide informative pathology materials like for pulmonology, cardiology and angiology. Therefore, ultrasound devices can be an ideal physical inspection method to be popularized.

On the other hand, professional medical knowledge can not be absent for analyzing the health condition data. Unlike blood glucose or blood pressure, ultrasound images should be interpreted by professional sonographers or doctors. In contrast to ultrasound devices' convenience in use, the analysis of ultrasound images requires a certain level of professional knowledge and experience due to its acoustic nature. The inherent drawback makes ultrasound images hard to be processed by smart diagnosis systems. Ahead of the diagnosis, an accurate and robust segmentation model can provide more precise information to the diagnosis process regarding to different organs. Also, the segmentation model can help reduce the burden of sonographers in current clinical practice. Besides pathological analysis, multi-organ segmentation algorithms can also be leveraged in other various fields. For example, a reliable multi-organ segmentation method can be incorporated into robot trajectory planning algorithm in the development of automatic ultrasound scanning system.

As recent advances in artificial intelligence (AI) are raising the competitiveness of computer-aided diagnosis (CAD) solutions against conventional diagnostic methods, the realization of end-to-end smart diagnosis is fast approaching. To realize end-to-end smart diagnosis, many researchers have made great contributions to medical and biological image segmentation methods. However, the field of ultrasound image segmentation still stays relatively less explored. Compared with other medical imaging modalities like computed tomography (CT) or magnetic resonance imaging (MRI), ultrasound images are harder to collect and analyze. Although there are a few researches revolving around ultrasound image segmentation, the importance of multi-organ segmentation is still ignored. On one hand, traditional morphological or statistical ultrasound image segmentation methods are usually time-consuming and of low flexibility. On the other hand, most of the AI based methods aim at only specific target organs, tissues, or anomalies. A robust segmentation method for multi-organ/tissue is still absent.

Segmentation of ultrasound images forms the cornerstone of end-to-end smart diagnosis systems. As what has been mentioned above, it can help with the navigation of ultrasound-scanning robots. And in one ultrasound image, we can run specified diagnosis programs on those well-cropped sub-regions according to different medical interests. In this work, we proposed and tested a neural network model that can perform end-to-end segmentation on abdominal ultrasound images targeting at 5 different anatomical structures (liver, kidney, vessels, gallbladder, and spleen). The primary contribution of this work is that to explore the field of multi-organ/tissue segmentation, compared with the previous research, this work takes two inherent features of ultrasound images into consideration: (1) different organs and tissues vary greatly in spatial sizes, (2) the anatomical structures inside human body form a relatively constant spatial relationship. Based on those two ideas, we propose a new image segmentation model combining feature pyramid network (FPN) and spatial recurrent neural network (SRNN). In the thesis, we will discuss why we use FPN to extract anatomical structures of different scales and how SRNN is implemented to extract the spatial context features in abdominal ultrasound images. The performance of our proposed model is evaluated both numerically and qualitatively. Evaluation results show the competitiveness of our proposed model, and the introduction of spatial contextual information has improved the performance of the pure feature pyramid network.

Keywords: Medical Imaging and Processing, Diagnostic Ultrasound, Image

Segmentation, Feature Pyramid Network.

List of Abbreviations

AI	Artificial intelligence
BRNN	Bidirectional recurrent neural network
CAD	Computer-aided diagnosis
CNN	Convolutional neural network
CT	Computed tomography
FPN	Feature pyramid network
GCCN	Global context convolutional network
MRI	Magnetic resonance imaging
RNN	Recurrent neural network
RPN	Region proposal network
SRNN	Spatial recurrent neural network
SSD	Single shot detector

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Chapter 1

Introduction

1.1 Background

After entering the 21st century, population aging is becoming a serious problem for many countries. Taking Japan as an example, according to Japan Statistical Yearbook 2023 released by the Minister of Internal Affairs and Communications [1], Japan's population aged 65 and over currently stands at just over 36.21 million, accounting for 28.9% of the total population. And this number is still on its way to the peak. One of the most challenging tasks brought by population aging is that senior citizens require regular and continuous health support or monitoring. To make sure senior citizens can get medical help timely and easily, besides medical facilities, diagnosing methods in the caring center or personal residence are also in need. Moreover, the popularity of personal medical inspection devices will bring convenience to other citizens as well. With the help of those devices, anyone in need can do simple physical checks to monitor their health condition just within their residence. For example, people with limited mobility or pregnant mothers will not bother traveling far to the hospital to get their regular physical check.

1.2 Research Motivation

The demand for real-time/convenient health monitoring requires remote or portable inspection devices to be popularized. In clinical practice, ultrasound imaging is one of the most commonly implemented imaging modalities. Because it is approachable, effective, informative and of low cost, ultrasound devices are easy to be implemented. Also benefiting from its non-invasive and non-radioactive nature, the operation of ultrasound devices is of low threshold (Figure 1.1). Medical ultrasound imaging requires an accurate delineation or segmentation of different anatomical structures for various purposes. For example, the doctors can make a naive health condition preview by assessing the organ size [2]. And in some surgeries, precise ultrasound image segmentation can provide guidance for the interventions [3]. However, in contrast

Figure 1.1: Portable Wireless Ultrasound Probe

to ultrasound devices' convenience in use, ultrasound images are hard to process because of low contrast, acoustic shadows, and speckles, to name a few [4]. It can be challenging even for experienced sonographers to detect the exact contour of anatomical structures, not to mention that it usually takes years of study and practice to train a qualified sonographer. Therefore, an automated and robust ultrasound image segmentation method is expected to help with locating and measuring important clinical information. Along the lines, we are developing a control algorithm for the robot arm to perform automatic ultrasound scans (see Figure 1.2). Because this system is expected to operate automatically without human intervention, an evaluation metric for the robot's movement is necessary. Besides resolution and clarity, the integrity of anatomical structures is important as well. To this end, a segmentation algorithm needs to be incorporated into the robot trajectory control system.

Ultrasound images derived from abdominal scanning is our primary research target. In clinical practice, doctors can leverage abdominal ultrasound scanning images to evaluate health conditions of various anatomical structures [2].

1.3 Research Contributions

In this research, we have designed and tested a new network model that can predict semantic masks on convex ultrasound images. Compared with previous researches of ultrasound image segmentation, this network model

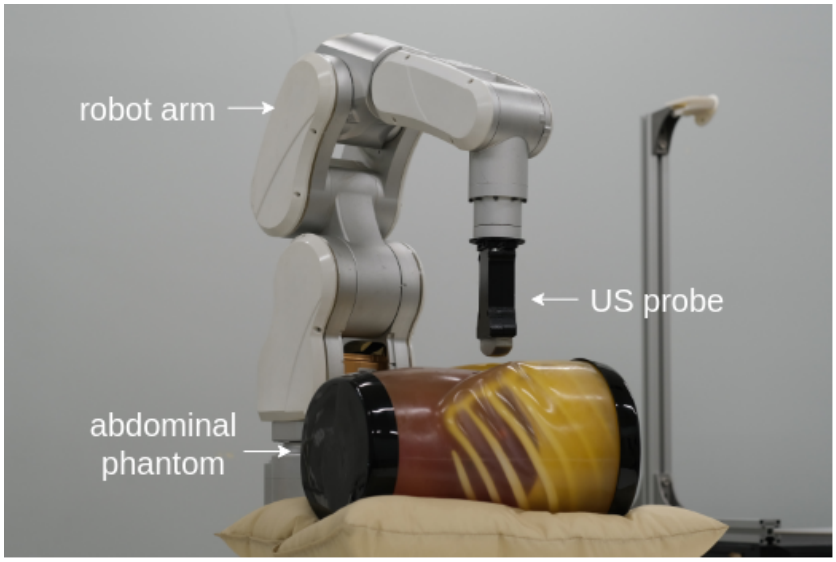


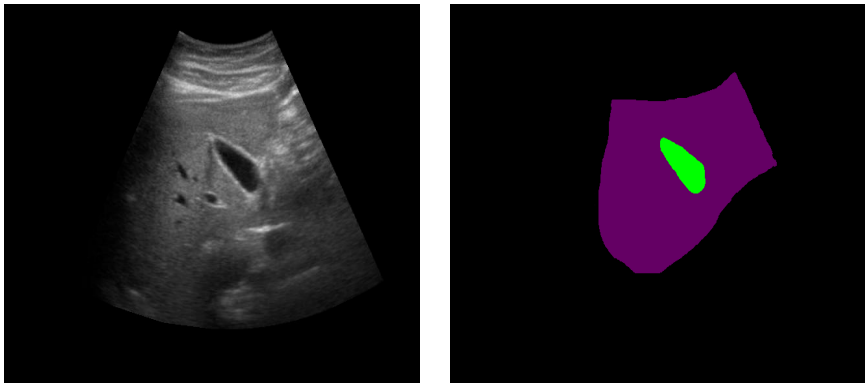
Figure 1.2: Our Robot-assisted Ultrasound Imaging System

can perform segmentation on several different anatomical structures simultaneously rather than focusing on a single target tissue. And to achieve this goal, two important inherent properties of ultrasound imaging are taken into consideration.

On one hand, different organs/tissues in abdominal part vary largely in shape and size. For example in Figure 1.3, liver(violet) is much larger than pancreas(green). To alleviate the influence of such class imbalance problem brought by the difference in scale, the network model is built in a FPN structure.

On the other hand, different anatomical structures usually form a constant spatial correlation pattern. For example in Figure 1.4, the spleen(pink) and kidney(yellow) are maintaining a similar spatial correlation in different ultrasound images.

Utilizing these two properties, the proposed model has managed to predict semantic masks for 5 different organs and tissues(liver, kidney, spleen, vessel, and gallbladder). Comparing with previous researches, this model realized true end-to-end ultrasound image segmentation aiming at multiple target structures using one single model. And this work can be generalized to other tasks where spatial context information can help a lot with analysis.



(a) Liver & Gallbladder (b) Mask Label

Figure 1.3: Scale Variance Example.

1.4 Thesis Outline

The rest of this thesis is organized as follows:

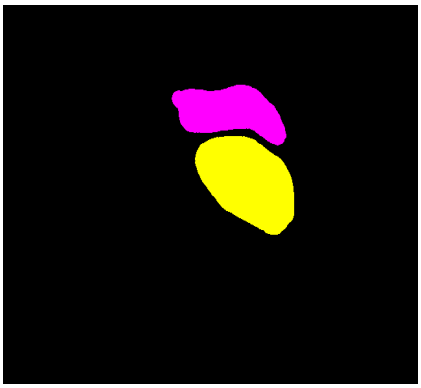
Chapter 2: In this chapter, we will list some of the related work revolving around neural networks and ultrasound image segmentation. Also, I will introduce their advantages and drawbacks and why this work can make a difference in the ultrasound image segmentation field.

Chapter 3: In chapter 3, we will introduce my methodology, including the details of the proposed network model.

Chapter 4: Explanation of the experiment settings and evaluation metric will be given in this chapter, then the performance of the proposed model will be analyzed both quantitatively and qualitatively.

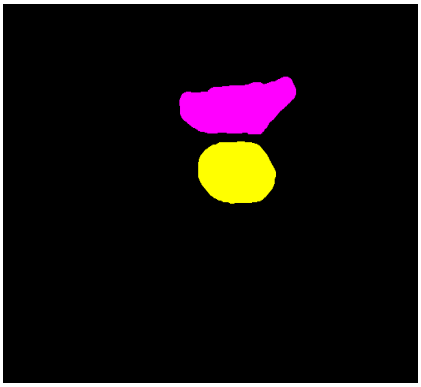
Chapter 5: Although the model has achieved encouraging results, improvements and explorations are still needed.

(a) Spleen & Pancreas(1)



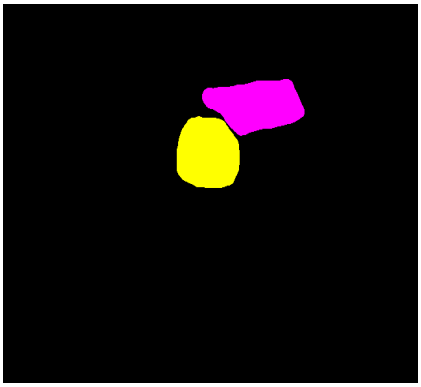
(b) Segmentation Mask(1)

(c) Spleen & Pancreas(2)



(d) Segmentation Mask(2)

(e) Spleen & Pancreas(3)



(f) Segmentation Mask(3)

Figure 1.4: Spatial Correlation Example.

Chapter 2

Related Work

2.1 Traditional Methods

Traditional ultrasound image segmentation methods usually focus on the detection of textures and boundaries based on morphological or statistical methods(Figure 2.1). Mishra et al. proposed an active contour solution using low pass filters and morphological operations to make a prediction of the cardiac contour [5]. Mignotte et al. developed a boundary estimation algorithm based on a Bayesian framework [6], where the estimation problem was formulated as an optimization algorithm to maximize the posterior possibility of being a boundary. Previously Mignotte's team used statistical external energy in a discrete activate contour for the segmentation of short-axis parasternal images [7], in which a shifted Rayleigh distribution was used to model gray-level statistics. Boukerroui et al. also proposed a Bayesian framework to conduct robust and adaptive region segmentation [8], taking the local class mean with a slow spatial variation into consideration to compensate for the non-uniformity of ultrasound echo signals. Those methods serve more as a reference for doctors than an independent diagnosis procedure.

Traditional ultrasound image segmentation is time-consuming and prone to irregular anatomical structure shapes because of the inherent physical defects of ultrasound. Low-contrast is suppressing the precision since it is hard to detect the contour. Also, deformation and low-clarity are making it even more challenging for traditional methods to predict correct segmentation. Those problems derive from ultrasound imaging's acoustic nature, so it is not easy to be removed in a computational way. Therefore in traditional ultrasound image segmentation researches, manual intervention operations are usually required, which is in contrast to our expectation of fully automated segmentation. Although the traditional segmentation methods may not be able to meet our expectations, their contributions towards the feature of organ contour or shape are often frequently introduced into convolutional

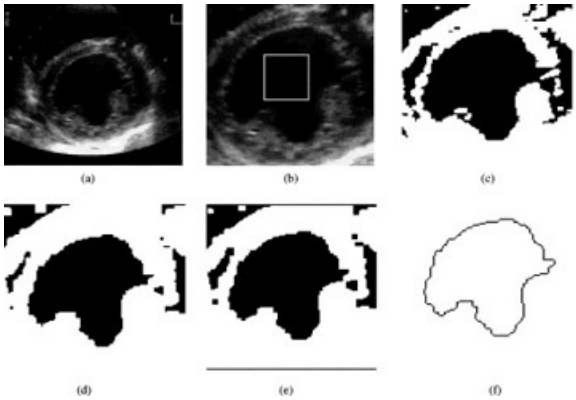


Figure 2.1: Traditional Segmentation Methods

neural network (CNN) based works as a piece of prior knowledge to improve the network model performance [9].

2.2 CNN Methods

In recent years, AI is showing great capacity in the field of image processing. More advanced algorithms and powerful computing devices are giving AI overwhelming competitiveness of handling various tasks. Compared with morphological and statistical methods, CNN based solutions are more powerful and flexible because of their strong nonlinear learning ability. There have been many applications of CNNs in medical image processing fields [10]. Zhang et al. conducted coarse and fine lymph node segmentation based on two series-connected fully convolutional neural networks [11]. Huang et al. proposed a modified deep residual U-Net model to predict the contour of abdominal organs and tissues [12]. They train their model initially on a tendon dataset, then fine-tune it on a breast tumor dataset. After getting a pre-trained model, they adapt the model to detect different anatomical structures using transfer learning. Li et al. proposed a male pelvic multi-organ segmentation method on transrectal ultrasound images [13]. In their research, a fully convolutional one-state (FCOS) object detector originally designed for generalized object detection is adapted for ultrasound image segmentation.

In the context of abdominal ultrasound image segmentation, most of the existing methods are targeted at specific organs or anomalies. Chen et al. designed a multi-scale and deep-supervised CNN architecture for kidney

image segmentation [14]. They implemented a multi-scale input pyramid structure to capture features at different scales, and developed a multi-output supervision module to enable the network to predict segmentation results from multi-scales. Huang et al. developed a detection algorithm for pulmonary nodules based on deep three-dimensional CNNs and ensemble learning [15]. However, the importance of multi-organ segmentation is still ignored. On one hand, the segmentation result can be used as a guide for the remote ultrasound scan system, which is the cornerstone of realizing an automatic remote diagnostic system. An automatic diagnostic system will reduce burdens on sonographers, enabling them to concentrate on the analysis of pathology. On the other hand, abdominal organ segmentation can also provide information on specific organs and tissues, which can be used to assist in the diagnosis of certain diseases. Therefore, in this paper, an abdominal multi-organ segmentation method is proposed. The contributed work can be listed as follows: (a) I proposed a multi-organ segmentation method based on an FPN structure; (b) I combined the FPN model with an SRNN module, which helps improve the performance significantly.

2.3 U-Net

U-Net is definitely one of the most popular network model for biological and medical image segmentation [16]. In the structure of U-Net, the downsampling part encodes the input image into feature representations at multiple different levels. Then the decoder part will upsample the feature maps and output a segmentation map. The main contribution of U-Net compared to other fully convolutional segmentation methods is that while features are upsampled and going deeper in the network, U-Net concatenates the higher resolution features from the down part with the upsampled features in the sense that the localization information is more precise in those layers. See Figure 2.2 for the structure of U-Net. Many researchers have built their own modified U-Net for various semantic segmentation tasks. For example, Oktay et al. [17] proposed a U-Net model with attention mechanism. They implemented attention gate units to trim features that are not relevant to the ongoing task in order to improve the segmentation performance without adding excessive computational complexity to the model. Matteo et al. [18] designed a Siam-U-Net for knee cartilage tracking. In that work, Matteo et al. extended the encoder of U-Net up to a parallel structure to extract the cross correlation depth wisely.

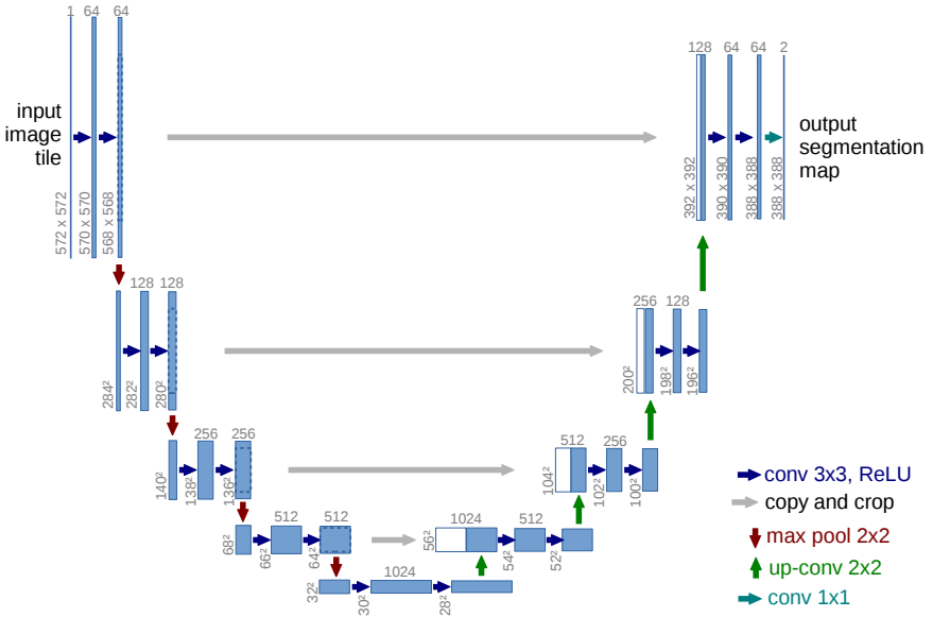
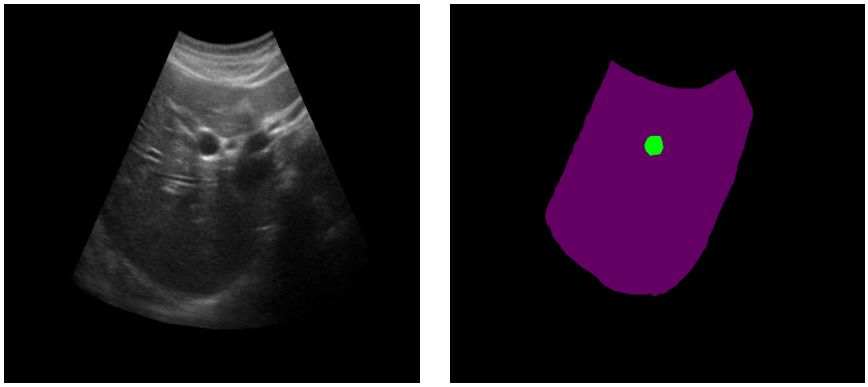


Figure 2.2: U-Net

Although U-Net has achieved great success in biological and medical image segmentation, in the context of abdominal multi organ segmentation, the performance of U-Net is limited by the class imbalance problem. In this work, despite that the total amount of instances may be almost equal in training, the relatively large organs and tissues occupy much more pixels in the ultrasound images. As shown in Figure 2.3, the violet part is the liver which occupies most of the pixels, and the green part is the gallbladder. This will make the algorithm classify as many pixels into the liver as possible, because a majority class has a much bigger influence on the final score than the minority class. Figure. 2.4 shows the segmentation result of an ultrasound image containing liver (violet) and kidney (yellow). Compared with the ground truth, the result tends to ignore the kidney to focus on drawing the true mask of the liver. There could be some solutions to this problem, such as adding different loss weights for different classes [19] or enlarging the weight of classification loss in the total loss calculation. However, to better leverage this property, we build our proposed network model upon an FPN structure [20].



(a) Ultrasound Image (b) Ground-truth

Figure 2.3: Example of Class Imbalance Problem

(a) Segmentation Result (b) Ground-truth

Figure 2.4: U-Net Segmentation Result

2.4 Spatial Context Extractors

One important property of ultrasound images is that the anatomical structures form a constant spatial relationship under the same scan pattern. For example, in the midsagittal plane scanning, the mesenteric artery is usually at the bottom of the liver, and the pancreas is located at the side of the liver. Experienced sonographers rely heavily on such spatial context information to locate the target organs. This prior knowledge inspired us to take spatial context information into consideration.

In CNN, convolution operation will gather the local features within a specific

receptive field, which is usually locally related with only a few pixels around the kernel center [21]. To extract global features in a long distance, there have been many variants of convolution algorithms. For example, Hamdi et al. [22] proposed a global context convolutional network (GCCN), which computes global features representing contextual information across image patches. In GCCN, global contextual features are defined as local maxima pixels with high visual sharpness in each patch. These features are then concatenated and utilized to augment the convolutional features. In the work of Duta et al. [23], they replaced the standard convolution kernel with contextual convolution kernels. Those kernels have a dilated shape so that they can extract information from spatially separated regions.

Compared with convolution-based methods, recurrent neural networks (RNNs) are more powerful to extract contextual information, especially direction-oriented context. Many studies have explored the utilization of RNNs to gather contextual information. Traditionally, RNNs are utilized to extract context from a sequence(sentence, speech, or video). For example, Schuster and Paliwal proposed a bidirectional recurrent neural network (BRNN) that passes both forward and backward across a time map to ensure the information is propagated across the entire timeline [24]. Tang et al. [25] designed a context-aware natural language generation model, which encodes the contexts into a continuous semantic representation and then decodes the semantic representation into text sequences with recurrent neural networks. When it comes to the context of spatial information, Graves and Schmidhuber proposed a multi-dimensional RNN to recognize handwriting [26]. Byeon et al. built a long short-term memory (LSTM) RNN structure for scene labeling [27]. In this work, SRNN is introduced.

Bell et al. proposed an object detection network structure called Inside-Outside Net (ION) [28]. Besides taking the information near an object's region of interest, the introduction of contextual information has improved the performance, for which a module of four directional RNNs is implemented. Fig. 2.5 shows the propagation of the RNNs. The structures are placed laterally across the feature maps, and move independently in four cardinal directions: right, left, down, and up. The outputs from the RNNs are then concatenated and computed as a feature map containing both local and global contextual information.

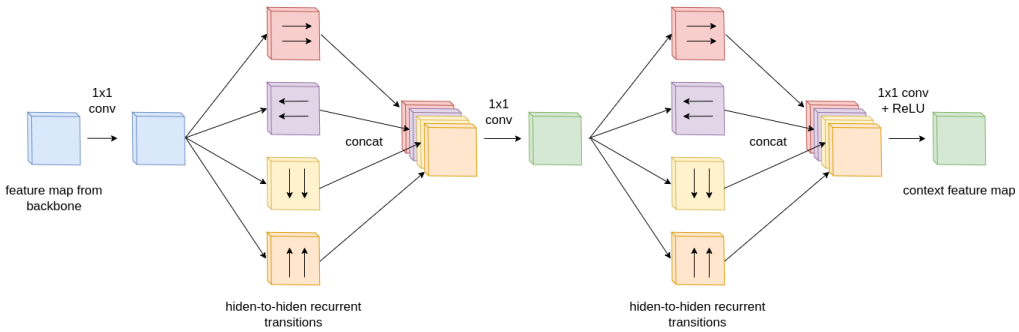


Figure 2.5: Spatial RNN Module

Chapter 3

Methodology

3.1 Feature Pyramid Network

In the abdominal section, different anatomical structures vary greatly in shape and size, which inspires us to leverage the FPN network structure. FPN is not an independent object detector by itself. It usually serves as a feature extractor for other detectors. Also FPN is not the exclusive name for any specific network model. We can build any FPN structure based on our own modified backbone targeting at different tasks.

The most important difference between FPN and its competitors is that FPN takes the feature maps from multiple layers of the encoder backbone as outputs rather than only from the deepest output [20]. Before FPN, there have been other kinds of network models following a pyramid structure. For example, the single shot detector (SSD) [29] is one of the first attempts at leveraging the feature pyramid hierarchy. The SSD reuse multi-scale feature maps from different layers in the forward pass. This pyramid network structure is scale-invariant in the sense that an object's scale changes with shifting its level in the feature pyramid. In other words, smaller objects are usually easier to be detected from smaller yet deeper feature maps, and vice versa. Compared with other pyramid network structures like SSD, FPN not only utilizes the relation between scale and layer depth but also uses a top-down pathway to construct higher-resolution layers from a semantic layer. This solves the problem that feature maps composed of low-level structures (closer to the original level) are too naive for accurate object detection. As the reconstructed layers are semantically strong, but the locations of objects are not precise after all the down-sampling and up-sampling, the authors then added lateral connections between reconstructed layers and the corresponding feature maps to help the decoder predict the locations better. Then the detector heads will make predictions on all the output layers. The overall structure of FPN is illustrated in Figure 3.1.

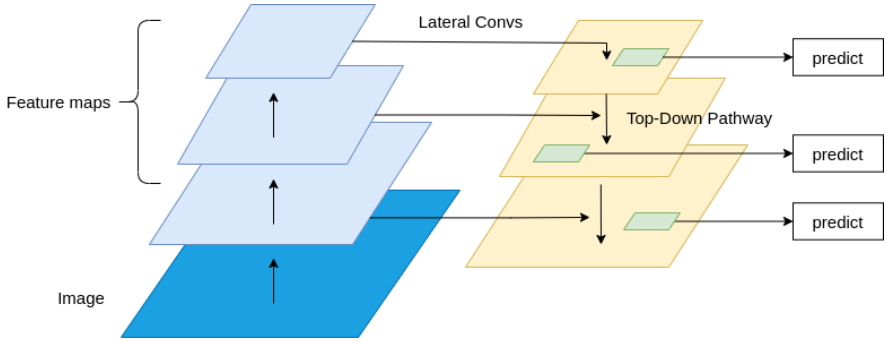


Figure 3.1: FPN Structure

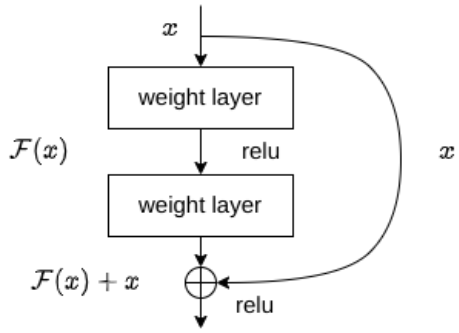


Figure 3.2: Shortcut Connection in ResNet

3.2 ResNet

We use ResNet [30] as the feature extractor backbone. ResNet is a deep residual learning framework to solve the degradation problem of deep networks. Theoretically, the prediction accuracy will not drop as the network goes deeper. However in programming practice, a common issue in deep learning is that deeper neural networks are harder to train. With the layers going deeper, accuracy would drop rapidly [31]. In other words, appending more layers to a suitably deep model will increase the training error. ResNet alleviates the degradation problem by reformulating some of the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions.

ResNet makes use of feedforward networks with "shortcut connections", which makes the network easier to optimize and able to gain extra accuracy from considerably increased depth. In the aforementioned paper, the authors let the shortcut connections simply perform identity mapping and produce

the sum of the output from original layers and the lateral layers as illustrated in Fig. 3.2, where x represents the input, $F(x)$ the abstract representation of the residual block, and relu the activation function. x is passed directly to the output and called "identity shortcut connection". They managed to train models with over 1000 layers by inserting the shortcut connection to the plain backbone. ResNet's capacity of extracting deep features made it possible in our work to combine both semantic and abstract information for analyzing ultrasound images.

3.3 SRNN Structure

The SRNN module follows the idea of the ION network structure. Figure. 3.3 shows how the RNNs extract the contextual information. A convolution operation is settled at the beginning of the precedure to replace the input-to-hidden translation. Then, four RNNs are propagated through the different directions mentioned above. The outputs from the RNNs are fused into an intermediate feature map. Until this step, each pixel contains the context information aiming at its four principal directions: right, left, up, and down. Then the model will conduct the same process again to extract global-level spatial context information. Finally, a feature map containing the overall context information is generated. For comparison, in the feature map on the left in Figure. 3.3, each pixel only contains information about itself and its neighbors (depending on the perspective field). After the first round of RNN propagation, the pixels get the context information from its 4 directions. Finally, RNNs propagate through the context-rich pixels to extract the full-directional context information. Therefore, the last feature map is globally context-rich.

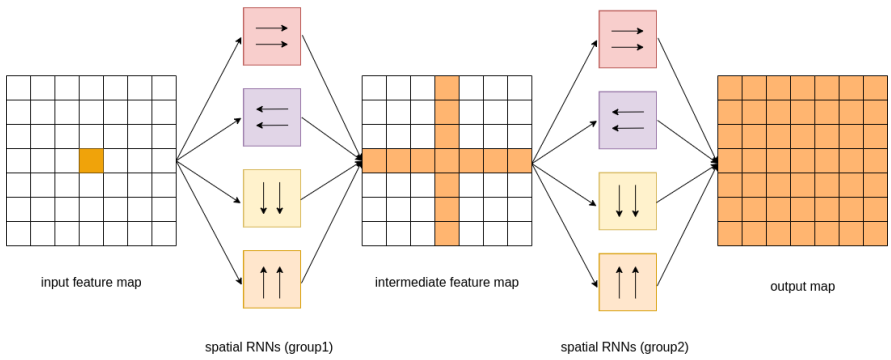


Figure 3.3: Illustration of the IRNN propagation

3.4 IRNN

An RNN is specialized for processing sequential data. The data fed into the input nodes is propagated through the hidden nodes, updating the internal states using past and present data. There are variants of RNN such as gated recurrent units [32], LSTM [33], and plain tanh RNNs. The RNN in this work follows the IRNN model in [34], because of its efficiency and simplicity of training. IRNN has a good performance for long-range data dependencies [28]. IRNN is composed of the rectified linear unit, and the recurrent weight matrix is initialized to the identity matrix. Therefore, gradients are propagated backward with full strength at initialization. We adapt four independent IRNNs that propagate through four different directions. Given below is the update function for the IRNN moving from left to right. The rest IRNNs follow a similar equation according to the propagation direction:

$$h_{i,j}^{right} \leftarrow max(W_{hh}^{right} h_{i,j-1}^{right} + h_{i,j}^{right}, 0), \tag{3.1}$$

where W is the hidden transition matrix and $h_{i,j}$ is the feature at pixel(i, j).

Each direction on independent rows or columns is computed in parallel, and the output from the IRNN is computed by concatenating the hidden state from the four directions at each spatial location.

3.5 Network Structure Overview

Figure 3.4 shows the overall structure of the proposed model. On the left side, we put a ResNet-101 backbone as the semantic feature extractor. The input image is propagated from bottom to top, with the network generating feature maps of lower resolution and richer semantic information. We define the layers producing feature maps of the same size as one stage. Then the output of the last layer of each stage to represent the output of the entire stage except the shallowest stage, because this high resolution layer is computationally consuming due to and of low semantic feature. Each of the blue cubes represents an output of the stage called {res2, res3, res4, res5}, respectively. The feature maps go separately through a 1x1 convolution layer and the SRNN module.

The green cubes represent the feature maps after the convolution operation, and the red cubes represent the context feature maps. The deep feature map is concatenated with the context feature map and compressed to reduce

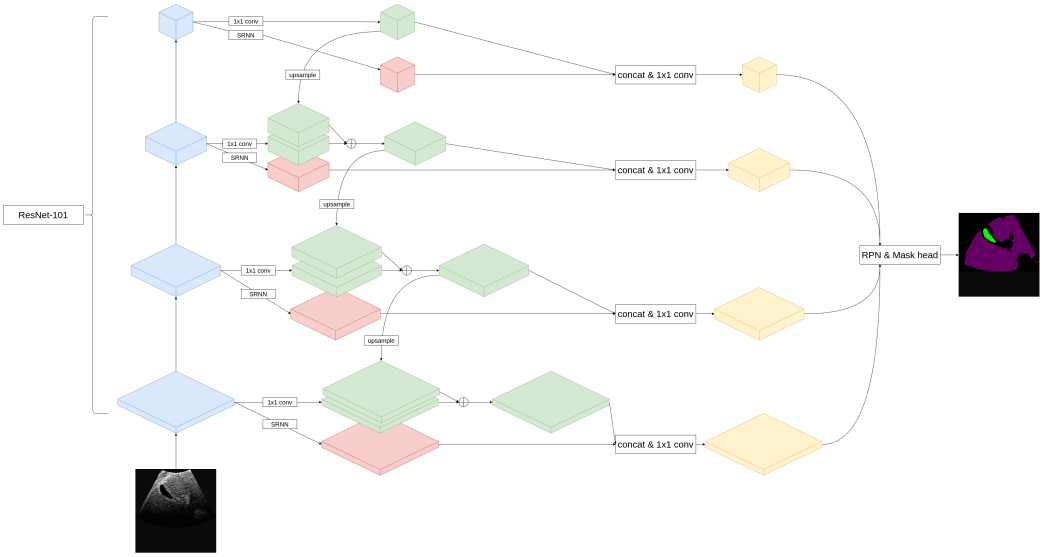


Figure 3.4: Proposed network structure

depth channels. The feature map from the upper layer, spatially coarser but semantically stronger, is upsampled by a scale factor of 2. (using interpolate function with nearest neighbor upsampling). The upsampled feature maps from the upper pyramid level and the feature map from the current pyramid level are added together as the new feature map to be concatenated with the spatial feature map. One thing worth noting is that using adding outperforms the solution of concatenating and dimensionality reducing.

The yellow cubes are the final outputs of the entire feature extractor. After extracting semantic and spatial features, these pyramid feature maps are then sent to region proposal network (RPN) [35] and region-based detectors (Fast R-CNN [35], Mask R-CNN [36]). Unlike the classic object detectors, the FPN attaches RPN and Fast R-CNN to each of the output layers. The parameters of the heads are shared across all feature pyramid levels for simplicity, but the accuracy is actually very close with or without sharing parameters ([20]). This is indirect proof that all the levels of the pyramid share similar semantic levels. After that, a DeepMask framework is used to generate masks. The structure of proposers and anchor/mask generators are omitted in the graph, since it is not our main interest.

Chapter 4

Experiment and Results

4.1 Dataset

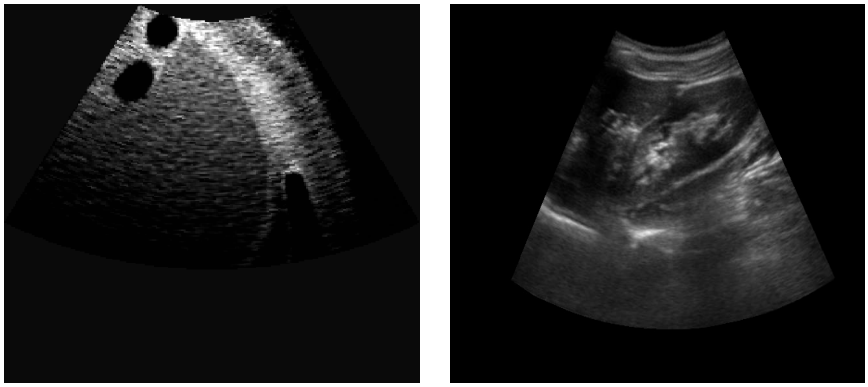
A dataset of high quality is one of the key factors to train a neural network. However, there are few open-source abdominal ultrasound image datasets. Most of the relevant researchers have not made their dataset public for the protection of patients' privacy. In this work, we use the dataset provided by Vitale et al. [37]. This dataset is released on Kaggle website and contains both artificial ultrasound images translated from CT images and images from real ultrasound scans(Figure. 4.1). There are 926 artificial ultrasound scans and 61 labeled real ultrasound scans, in which we can have the annotations of the liver, kidney, gallbladder, spleen, and vessels. Different organs are assigned segmentation masks of different colors. Table 4.1 shows the name of the anatomical structures and the corresponding instance number. We mixed and separated the dataset into 3 subsets: 798 images for training, 100 for testing, and 100 for validation. This dataset also contains some unlabeled real ultrasound scan images. However, it has to be labeled by professionals. In future work, we will consider seeking collaborations with medical facilities to label those images and get new sources of high-quality ultrasound images.

Table 4.1: Dataset

Name	Liver	Kidney	Gallbladder	Vessels	Spleen
Number	591	377	219	289	172
Color	violet	yellow	green	red	pink

4.2 Detectron2

In 2019, the Facebook research team released a powerful object detection tool called *detectron2* containing many network architectures and training



(a) Artificial Ultrasound Image (b) Real Ultrasound Image

Figure 4.1: Dataset Content Example

tools [38]. This great work allows users to train the given networks on their specific tasks or the users can build their own structure efficiently by using the encapsulated models. We build the backbone framework based on the implementation of FPN in *detectron2*. And we develop our SRNN structure inserted into the FPN framework as a new context feature extractor. The standardized RPN, ROI, Fast R-CNN, and Mask R-CNN heads are attached after the feature extractor as the proposal generators. Specifically, the output feature maps are from {res2, res3, res4, res5} of the ResNet layers. The size of the anchor generators are set to $32 \times 32, 64 \times 64, 128 \times 128$, and 256×256. For each feature map, FPN gives 1000 proposals. The ROI box head follows the structure of Fast R-CNN with 2 fully convolutional layers and 7×7 pooler resolution. The Mask R-CNN head has 4 convolutional layers and a pooler resolution of 14×14. The ROI heads score threshold is set to 0.5 for both box and mask heads. We have made some modifications to the model, enabling it to run under the *detectron2* framework. For example, we use a small learning rate to ensure there will not be NaN and infinity scores in the final result and reduce the ROI head batch size from 512 to 128, which is computationally efficient while the accuracy is nearly the same.

4.3 Loss Functions

Multiple loss functions are included in our training procedure, some of which are listed in this section.

4.3.1 Objectness loss

For detection of the object appearance, the binary cross-entropy loss is used in the RPN head. This loss is only for the classification of object and background. In RPN, this loss will be computed on the objectness logits feature map and the ground truth objectness logits. If the pixel is from some target object, it will be marked as "1", otherwise "0". The formulation is:

$$L_{obj} = -\frac{1}{n}\sum_{i=1}^{n} y_i \log\left(p(y_i)\right) + (1 - y_i)\log\left(1 - p(y_i)\right), \tag{4.1}$$

where y is the label ("1" for foreground, "0" for background), and $p(y)$ is the predicted probability for all the grid points in the feature map.

4.3.2 Anchor and bounding box loss

Both RPN and ROI(Box) heads use a smooth l1 loss for the proposed anchors and bounding boxes. The anchors and bounding boxes are represented as a tensor of length 4: (x, y, w, h), namely the x, y coordinates and width/height of the anchor or bounding box. Then with the ground truth information, 4 deltas (d_x, d_y, d_w, d_h) are calculated by

$$\begin{aligned} d_x &= (g_x - p_x) \\ d_y &= (g_y - p_y) \\ d_w &= \log\left(g_w/p_w\right) \\ d_h &= \log\left(g_h/p_g\right) \end{aligned} \tag{4.2}$$

where g represents the ground truth and p stand for the predicted anchor or bounding box. The deltas will be stacked together to compute the smooth l1 loss, given by

$$L_1^{smooth}(x) = \begin{cases} 0.5x^2 & if \quad |x| < \beta \\ |x| - 0.5 * \beta & otherwise \end{cases} \tag{4.3}$$

where β is a pre-defined smooth parameter.

4.3.3 Classification loss

Softmax cross entropy loss is calculated for all the foreground and background prediction scores:

$$L_{CE} = -\sum_{i=1}^{n} y_i log(p_i) \tag{4.4}$$

where y_i is the true label and p_i is the softmax probability for the ith class.

4.3.4 Mask loss

The mask loss is defined as the average binary cross-entropy loss. Eq. 4.5 computes the mask loss for the kth class:

$$L_{mask} = -\frac{1}{m^2} \sum_{1 \leq i,j \leq m} [y_{i,j} \log \hat{y}^k_{i,j} + (1 - y_{i,j}) \log(1 - \hat{y}^k_{i,j})] \tag{4.5}$$

where $y_{i,j}$ is the label of a cell(i,j) in the true mask for the region of size $m \times m$, and $\hat{y}^k_{i,j}$ represents the value of the same cell in the predicted mask.

4.4 Experiment Setup

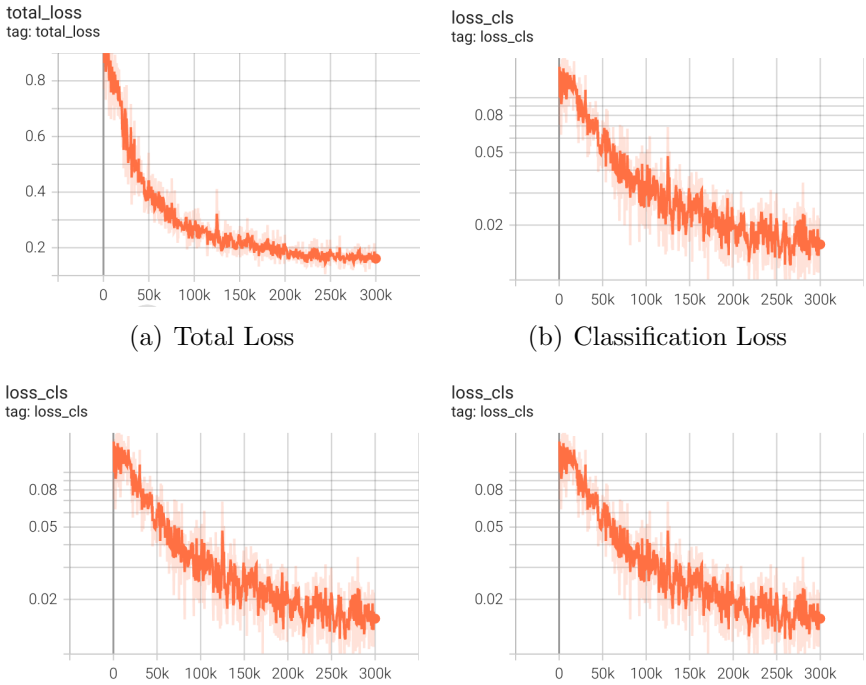
Our experiment builds upon *detectron2* framework in PyTorch environment. We modified the original FPN in *detectron2* via adding the SRNN module. We train the model on a single GPU (RTX3080). The batch size is set to 1 since this GPU has relatively a limited memory and the dataset is again relatively small. The model is trained for 300k epochs, taking around 30 hours to converge. The initial learning rate is set to 0.0025, but the SRNN module is prone to this learning rate so that the final prediction scores of the ROI heads may return NaN/inf after the SRNN module is inserted. Therefore, we opted to use a small learning rate of 0.00025 as the initial setting. We also tried to explore deeper layers in the backbone, adding 2 extra pyramid layers on the top of the backbone, but the accuracy fails to increase. Fig. 4.2 shows the converge curve of the proposed model, namely the loss curve of (a) total loss, (b) classification loss, (c) bounding box region loss, (d) and mask loss.

4.5 Results

4.5.1 Quantitative Result

The performance of the trained model is evaluated by the following dice coefficient:

$$D = \frac{2|X \cap Y|}{|X| + |Y|} \tag{4.6}$$



(a) Total Loss (b) Classification Loss

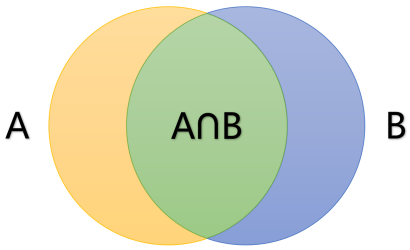
(c) Bounding-box Region Loss (d) Mask Loss

Figure 4.2: Loss Curves

The dice coefficient(illustration see in Figure 4.3) is twice the number of elements common to two sets X and Y, divided by the sum of the number of elements in each set. In our work, X and Y are the predicted classification map and the ground truth. Therefore, the numerator is regarded as the intersection pixels of the predicted mask and the ground truth, and the denominator the sum of mask pixels in both. Considering we have 5 object classes (background not included), the coefficient score is computed separately and then averaged as the final score. As there might be no appearance of certain classes, we added a smoothing parameter ϵ to avoid zeros in the denominator. The modified equation is given in (4.7), where n is the number of classes.

$$D = \frac{\sum_{i=1}^{n} \frac{2|X_i \cap Y_i| + \epsilon}{|X_i| + |Y_i| + \epsilon}}{n} \tag{4.7}$$

There are few similar researches surrounding abdominal multi-organ segmentation. Neither any relevant benchmark or competitionexists. Therefore, we separately pick some comparable results from different researches aiming at single organ segmentation. Respectively, the segmentation result of liver is



$$Dice = \frac{2|A \cap B|}{|A| + |B|} \qquad Dice = \frac{\sum_{i=1}^{n} \frac{2|A_i \cap B_i| + \epsilon}{|A_i| + |B_i| + \epsilon}}{n}$$

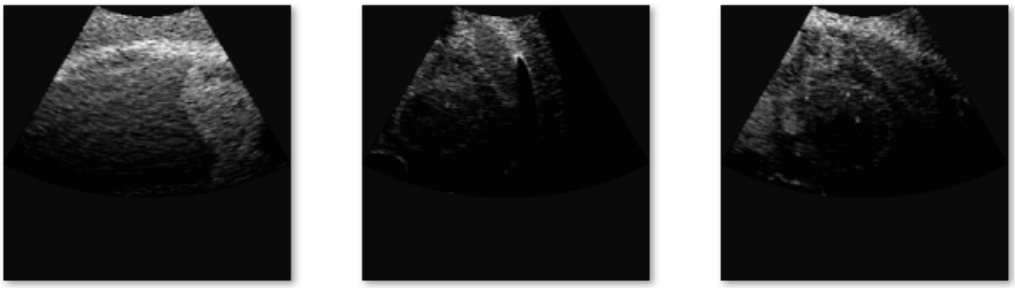
Figure 4.3: Dice Coefficient

compared with the work [39] kidney [40], their result is taken into comparison as well. And the segmentation performance of gallbladder and spleen are compared with [41] and [42]. The numeric result may not seem encouraging compared with those well aimed researches. On one hand, the segmentation performance is limited by our lack of high-quality data. For example, in the work of Yuan et al. [42], they trained their model on 420 good quality 2D spleen ultrasound images. In our research, we have only 172 instances of training spleen training samples, not to mention that most of the ultrasound images are pseudo ultrasound images interpreted from CT image. See Figure 4.4 for the difference. On the other hand, the specifically targeted researches usually introduced some prior knowledge into their segmentation algorithm like the detection of boundaries. Meanwhile, we trained a pure FPN model for comparison to demonstrate the improvement brought by SRNN. Table 4.2 shows the dice score of each class, where we can see that the improvement of performance by SRNN is significant. The proposed model outperformed the pure FPN model.

Table 4.2: Evaluation Result

Organ/Tissue	Related Work	FPN	FPN+SRNN
Liver	0.821 [39]	0.907	0.924
Kidney	0.5 [40]	0.806	0.836
Gallbladder	0.893 [41]	0.799	0.815
Vessels	–	0.801	0.825
Spleen	0.93 [42]	0.810	0.859
Average	–	0.840	0.865

(a) High-quality Dataset



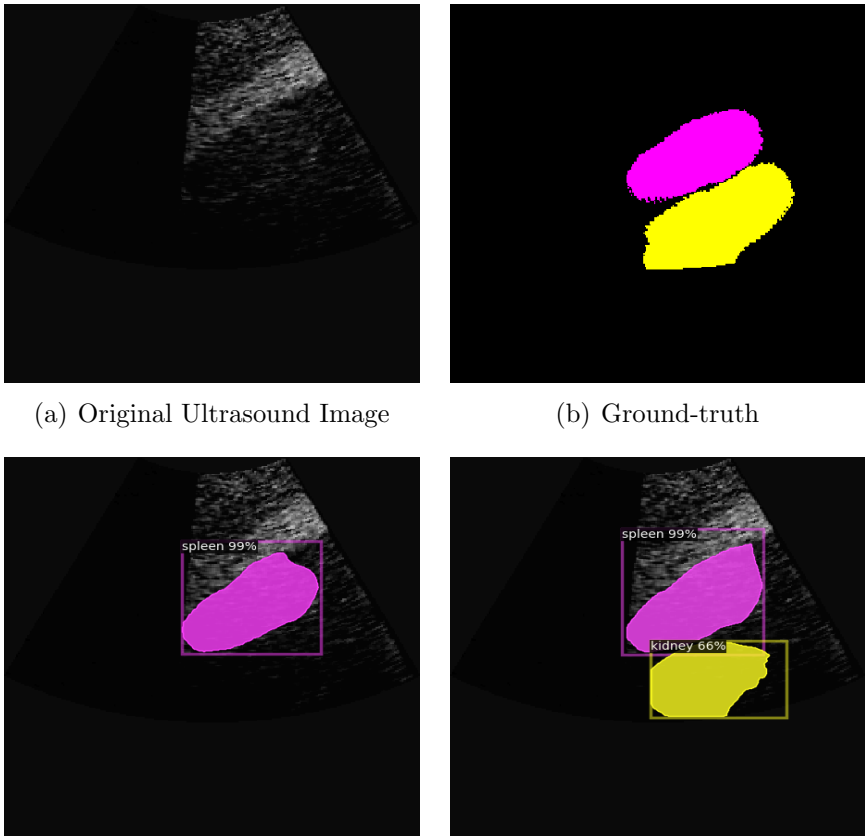
(b) Our Dataset

Figure 4.4: Dataset Difference

4.5.2 Qualitative Result

We have tested the proposed model on the artificial and real ultrasound images from the evaluation data. Fig. 4.5 shows an example of the semantic segmentation on ultrasound image. (a) is the original ultrasound image, (b) is the corresponding ground truth. (c) and (d) are the segmentation result generated by the pure FPN and our proposed model. We can see that the proposed model outperforms the pure FPN.

Furthermore, our proposed model was tested on the ultrasound images collected manually from an abdominal phantom in our laboratory. Fig. 4.6 shows an outstanding performance of our model: (a) is a ultrasound image collected from the phantom, (b) is the semantic masks generated. One thing worh noting is that the green mask (corresponding to gallbladder) is actually mispredicted. It has a trust score of 73%. Trust threshold is a hyper-parameter in training to filter out low-scored bounding boxes predicted by the Fast R-CNN component of the model during inference/test time. Basically, any prediction with a confidence score above the threshold value

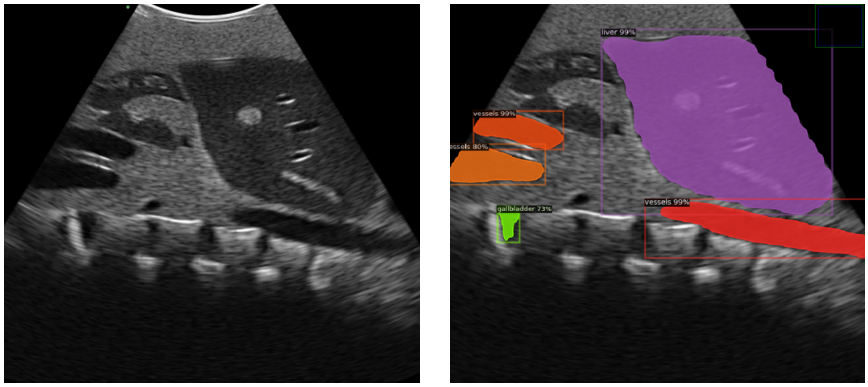


(a) Original Ultrasound Image (b) Ground-truth

(c) FPN Result (d) Proposed Model Result

Figure 4.5: Test Results

is kept, and the remaining are discarded. After several experiments, we found that changing this hyper parameter will not increase the accuracy score numerically, but the network does show a different trend of prediction.



(a) In Vitro Ultrasound Image (b) Proposed Model Result

Figure 4.6: Robot-assisted Ultrasound Image Capture/Segmentation

Chapter 5

Conclusion and Future Work

5.1 Conclusion

In this research, We proposed an FPN based multi-organ/tissue segmentation method combined with the utilization of SRNN. From the experimental results, we can see that the introduction of spatial context information has improved the performance of the original FPN model both in quantitative and qualitative comparison. And our model is competitive even compared with those well-targeted researches. The success of this work lays a solid foundation for feature extension like the development of fully automated ultrasound scan system and end-to-end abdominal ultrasound diagnosis solution. The findings of this work would also benefit from further research including different scan patterns, since a prior knowledge of the US scan pattern would help add more precise spatial context information.

5.2 Future Plan

The future plan of this research is expected to contain the following work.

US scan pattern recognition: In clinical US scan practice, different scan patterns are performed according to different medical interest like midsagittal plane scan, right costal interspace section scan, axial section scan, etc. Spatial correlations of anatomical structures will change with different patterns. Therefore, a prior knowledge of the US image's scan pattern would help add more precise spatial context information.

A high-quality dataset: We are seeking the chance of cooperation in building an open-source medical US image dataset that can accelerate our timeline of bringing an AI diagnostic system into reality.

Robot-assisted ultrasound imaging: The ultimate goal of our project is to build a control algorithm for robot arm to perform automatic US scan.

Therefore, the improved segmentation algorithm will be introduced as the evaluation metric of the control system performance.

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